

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		${}^{92}_{36}\text{Kr}$		2		2	2	
		(ii)		Moderator	1			1		
		(iii)		3		1		1		
	(b)	(i)		To absorb (excess) neutrons (1) so a mean of 1 neutron from each fission produces further fission / so that for one neutron in there is one neutron out (causing further fission) / prevent an uncontrollable chain reaction (1)	2			2		
		(ii)		The control rods may not be able to absorb {sufficient/enough} neutrons (to control the reaction) (1) and so the reactor may so overheat / or meltdown / uncontrollable chain reaction (1) Do not accept: explode		2		2		
	(c)	(i)	I	700 million years (1)		1		1	1	
			II	700 million years (1) ecf		1		1	1	
		(ii)		50 (1) 12.5 (1) Do not accept 13		2		2	2	
				Question 3 total	3	9	0	12	6	0